

Which of the following are true?

Let $f(n) = 4n(n + 2 \log_2 n^2) + e^{-n} + 8 \sin\left(\frac{2\pi n}{256}\right)$

We can show that $f(n) = o(g(n))$ if:

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$$

We can show that $f(n) = O(g(n))$ if:

$$f(n) = o(n \log_2 n)$$

This is false since $f(n)$ grows faster than $g(n)$

- for $n = 100$:

$$\frac{f(100)}{g(100)} = \frac{\sim 41605}{200} = \sim 208$$

- for $n = 10000$:

$$\frac{f(10000)}{g(10000)} = \frac{\sim 400320003}{132877} = \sim 3012$$

$$f(n) = o(2^n)$$

This is true since $f(n)$ grows more slowly than $g(n)$

$$f(n) = o(n^4)$$

This is true since $f(n)$ grows more slowly than $g(n)$

$$f(n) = O(n \log_2 n^n)$$

$$n \log_2 n^n = n^2 \log_2 n$$

This is false since for every $n > 2$ then $f(n)$ grows faster than $g(n)$

$$f(n) = O(2^n + 1)$$

This is true because $2^n + 1 \geq n^2$

 M_2 modified

$$f(n) = (2k + 1)(n + 1), \text{ where } k = \text{ceil}(\log_2 n)$$

- $f(8) = 63$ my turing machine does the check in 44 steps
- $f(10) = 99$ my turing machine does the check in 70 steps
- $f(30) = 341$ my turing machine does the check in 262 steps
- $f(60) = 793$ my turing machine does the check in 634 steps
- $f(100) = 1515$ my turing machine does the check in 1264 steps

“a reasonable time”**Brute force**

We run Dijkstra's algorithm for all permutations of every transportation request and if there are no feasible solutions we return false otherwise we return the shortest path. This would be a very slow algorithm since we would have to run Dijkstra's algorithm $n!$ times.

Better algorithm